

A GAI-Based Haptic Transmission Architecture for Extending Headset Lifespan in Haptic-Enhanced XR

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Abstract—Moving computing components from headsets to cloud servers is a promising approach to increasing headsets' lifespan and comfort for Extended Reality (XR) users. However, latency is an unavoidable challenge for XR services due to long-distance transmission, especially when haptic feedback (usually requires 1 ms latency) is involved. To address this challenge, we leverage the Latent Diffusion Model (LDM) to propose a novel transmission architecture, which can accurately generate the future potential haptic feedback from current or previous video frames. Moreover, we also introduce an acceleration architecture to accelerate the haptic feedback generation process. The simulations indicate that the lifespan of headsets can be tripled by moving computing resources to the cloud. In addition, our proposed architecture can accurately generate future potential haptic feedback at least 170 ms before contact, which can satisfy the 1 ms latency requirement of haptic feedback.

Index Terms—Haptic, green communication, generative artificial intelligence, edge networks

I. INTRODUCTION

In recent years, XR has delivered highly realistic visual experiences through 4K resolution and 360-degree video displays. With the involvement of haptic feedback, XR services will evolve into haptic-enhanced XR services, providing more immersive experiences to end users [1]. However, the limited computing power of mobile XR devices, e.g., headsets has become the bottleneck in the development of haptic-enhanced XR services [2].

Moving computing tasks from headsets to the cloud allows the headset to significantly reduce its energy consumption by retaining only communication and display components [3]. However, rendering in the cloud introduces long-distance latency, making it extremely challenging to meet the 1 ms latency requirement for haptic feedback [4]. Moreover, haptic feedback is vulnerable to network fluctuations and environmental interference, which means data loss or retransmissions will significantly affect the real-time performance of haptic feedback [5].

Humans can imagine potential haptic feedback simply by looking at a scene without actually making contact. For instance, simply seeing a fuzzy fabric and a glass surface allows us to imagine how each would feel to touch. Inspired by this, if the future potential haptic feedback can be generated before the contact occurs, the long-distance latency could be addressed. To realize this idea, we propose a haptic transmission architecture to generate haptic feedback using LDM [6]

in generative artificial intelligence (GAI), which generates the future potential haptic feedback before the contact occurs and sends the generated haptic feedback to users in advance to reduce the haptic end-to-end latency. The main contributions are as follows:

- Inspired by the complementary haptic feedback of the human brain, we employ LDM to generate future potential haptic feedback based on the current and previous video frames before the actual contact occurs.
- To speed up the haptic feedback generation process, we introduce an acceleration architecture that caches approximately unchanged features during the future sampling steps of the LDM to reduce the computational load.
- Through simulations, the lifespan of headsets can be tripled. And we are able to deliver haptic feedback to users 170 ms before actual contact. We are able to guarantee close to 100 percent haptic feedback transmission success rate even when the traffic load approaches 100%.

The remainder of this paper is organized as follows: Section II discusses related works. Section III shows the details of how the problem is modeled. In Section IV, we demonstrate the details of the proposed architecture. Section V illustrates the simulation settings and results. The paper is concluded in Section VI.

II. RELATED WORK

Since haptic feedback is highly sensitive to latency, various efforts have been made to address this challenge. For example, authors in [7] multiplexed multiple access schemes for different traffic flows, enabling haptic feedback with a latency of 2 ms and a reliability of 99.999%. In recent years, Long Short-Term Memory (LSTM) can be used in latency-sensitive real-time tasks [8], Wang et al. [9] utilized LSTM to generate future haptic feedback based on the current haptic feedback. Then the generated haptic feedback is sent out in advance to offset their transmission latency. However, this approach fails to consider the time required for both transmission and generation of haptic feedback, making it hard to satisfy the 1 ms latency requirement. In addition, efficient and high compression ratio haptic encoders are proposed in work [10]–[12], they can also effectively reduce the latency of haptic feedback by reducing encoding time and compressing haptic feedback into smaller packets. Authors in [13] propose a new mobile crowdsensing

architecture by combining C-RAN with D2D to meet the latency requirements demanded by 5G networks.

Additionally, there are several existing methods that do not transmit haptic feedback from a remote source, but derive it directly from the content of other modalities. Xue et al. [14] have investigated the interaction between video and haptic feedback, and estimated the kinesthetic position of haptic control with gaze information. Authors in [15] constructed a cross-modal connection between vision and haptic by leveraging Generative Adversarial Networks (GANs). The proposed approach can synthesize reasonable haptic feedback from visual inputs, and infer the interaction mode of an object based on the haptic feedback. Yang et al. [16] utilized a diffusion model to generate haptic feedback, and it can generate higher accuracy haptic feedback compared to GANs. However, these approaches are insufficient for solving the long-distance latency problem because they can only generate the concurrent related haptic feedback from current video frames, i.e., the related haptic feedback can only be generated when the contact occurs in the video. Therefore, the requirement for 1 ms latency remains unresolved due to the significant transmission and generation time of haptic feedback. Consequently, it is essential to utilize the video frames related to the expected contact to generate future potential haptic feedback before the interaction occurs. In this way, the haptic feedback could be delivered to end users prior to meeting its 1 ms latency requirement.

III. SYSTEM MODEL

In this section, we present the details of mathematical modeling for the proposed transmission architecture.

A. Network Model

We adopt haptic-enhanced XR services with haptic feedback, audio, and video content as an example. Where users request haptic-enhanced XR content from cloud servers. Our proposed architecture is deployed on edge servers for generating potential haptic feedback based on XR content.

The transmission of haptic-enhanced XR services is illustrated in Fig. 1. We use $S = \{S_c, S_e\}$ to represent the set of servers, where S_c stands for the cloud servers, and S_e are the edge servers. The haptic-enhanced XR services are defined as $C_n = \{H_n, V_n, A_n\}_{n=1}^N$ where N is the total number of haptic-enhanced XR services, and H_n, V_n, A_n are the haptic feedback, video segment, and audio segment respectively. H_n, V_n and A_n can be further expressed as $H_n = \{h_n^{t_1}, h_n^{t_2}, \dots, h_n^{t_i}, \dots, h_n^{t_{con}}\}$, $V_n = \{f_n^{t_1}, f_n^{t_2}, \dots, f_n^{t_i}, \dots, f_n^{t_{con}}\}$ and $A_n = \{a_n^{t_1}, a_n^{t_2}, \dots, a_n^{t_i}, \dots, a_n^{t_{con}}\}$ where t_{con} is the contact time, $h_n^{t_i}, f_n^{t_i}$ and $a_n^{t_i}$ are the haptic feedback, video frame, and audio clip at time t_i . Specifically, a touch action can be categorized into two phases: the pre-contact phase (moving toward the object) and the contact phase (interacting with the object). Hence, in the pre-contact phase, there is no haptic feedback since no contact occurs. So $h_n^{t_i}$ can be considered zero if t_i is not equal to t_{con} . Thus, the C_n can be rewrote as:

$$C_n = \{(f_n^{t_1}, a_n^{t_1}), (f_n^{t_2}, a_n^{t_2}), \dots, (f_n^{t_i}, a_n^{t_i}), \dots, (h_n^{t_{con}}, f_n^{t_{con}}, a_n^{t_{con}})\}. \quad (1)$$

We suppose the user receives a haptic feedback $h_n^{t_{con}}$ at time t_r , then the following equation holds:

$$t_r - t_{con} < l_{max} \quad (2)$$

where l_{max} is the maximum latency required by the haptic feedback.

B. Latent Diffusion Models

LDMs perform the diffusion step by compressing the data into the latent space, thus drastically reducing the generation time compared to Diffusion Models (DMs). LDMs have more flexible conditional inputs than other GAI models and have more powerful generative capabilities to meet our need to generate high-accuracy haptic feedback based on video frames.

We then discuss the image generation process as an example. The image I is first compressed to the latent space z_0 by the encoder $\mathcal{E}$, and the dimension of z_0 is much smaller than I . Subsequently, the z_s is obtained by adding noise to z_0 after s -steps:

$$z_s = \sqrt{\bar{\alpha}_s} z_0 + \sqrt{1 - \bar{\alpha}_s} \xi_s, \quad \xi_s \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (3)$$

where $\bar{\alpha}_s = (\alpha_0 \times \alpha_1 \times \dots \times \alpha_{s-1} \times \alpha_s)^2$, α_s is the factor of the noise addition process at step s , and ξ_s is the added noise at step s . Then the denoising network ϵ_θ can be trained by the following equation:

$$\nabla_\theta \|\xi_s - \epsilon_\theta(z_s, \mathbf{E}_C(\mathcal{P}), s)\|^2 \quad (4)$$

where $\mathbf{E}_C$ is the conditional encoder, the conditional input is denoted by $\mathcal{P}$, and $\mathbf{E}_C(\mathcal{P})$ is provided to the denoising network via cross-attention mechanism to control the denoising process and the generation of the final image. In the sampling phase, noise reduction is performed on the basis of the noise $z_q = \xi_q$ and $\xi_q \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ which can be described as follow:

$$z_{q-1} = z_q - \epsilon_\theta(z_q, \mathbf{E}_C(\mathcal{P}), q). \quad (5)$$

z_0 is passed through the encoder $\mathcal{D}$ and reverted to the dimension of I after q steps of noise reduction. In this way, an image is generated from the randomly sampled noise ξ_n .

C. Haptic Feedback Transmission Model

The transmission of haptic feedback is divided into two processes, the generation of future potential haptic feedback and the transmission of the generated haptic feedback to the user. At the generation process, the required generation time is l_g (usually at hundred ms level). Since the haptic feedback is represented by haptic image, it requires processing time to convert the haptic image H_n into actual haptic feedback, and the processing time is l_o (usually at ms level). During the transmission process, transmission latency to send the haptic feedback H_n from the edge server to the user side is denoted as l_c (usually at tens of ms level). Since the sum of l_c, l_o and l_g is significantly larger than the maximum end-to-end latency l_{max} (usually at 1 ms) required by the haptic feedback, it is essential to generate the future potential haptic feedback before the actual physical contact occurs. We define the time that generates future potential haptic feedback before the contact as

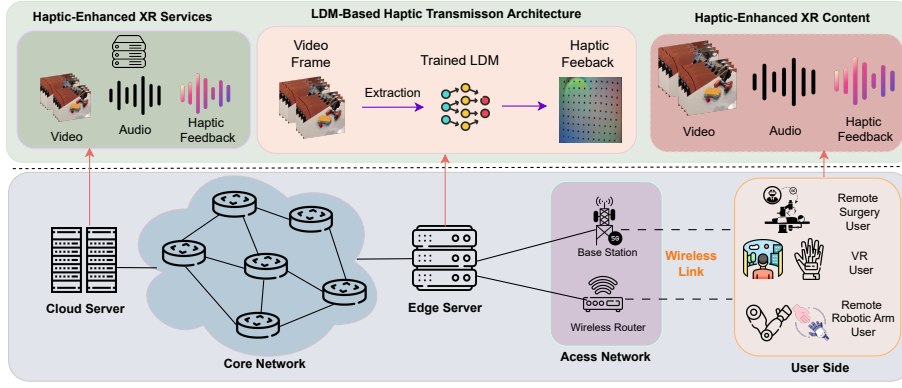


Fig. 1: The proposed GAI-based transmission architecture for haptic-enhanced XR services.

T_p . To provide natural haptic experience for users, T_p should satisfy the following condition:

$$T_p \geq l_g + l_o + l_c - l_{max}. \quad (6)$$

Therefore, to better ensure the haptic feedback meets end-to-end latency requirement l_{max} , we can either increase T_p or reduce l_g , because of the inherent difficulty in reducing transmission latency l_c .

According to Equation (1), the process of touching is divided into two phases, and the corresponding video frames V_n can be divided into two categories: pre-contact frames $\{f_n^{t_1}, f_n^{t_2}, \dots, f_n^{t_i}, \dots, f_n^{t_{con}-1}\}$, and first contact frame $f_n^{t_{con}}$. Since t_i satisfies the equation: $t_i = t_{con} - T_p$, Equation (6) can be rewrote as follows:

$$t_i \leq t_{con} - l_g - l_o - l_c + l_{max}. \quad (7)$$

This equation states the maximum case of t_i , that we cannot use video frames that are too close to the moment t_{con} to generate haptic feedback. Moreover, the video frames cannot be too far away from the t_{con} since they may not carry enough information to accurately generate the haptic feedback that may occur. We assume that the earliest and latest times of t_i are t_{min} and t_{max} respectively, then the range for T_p is $t_{con} - t_{max}$ to $t_{con} - t_{min}$. Thus the set of video frames we use to generate the haptic feedback $h_n^{t_{con}}$ is the set $\chi = \{f_n^{t_{min}}, f_n^{t_{min}+1}, \dots, f_n^{t_i}, \dots, f_n^{t_{max}}\}$. Then we extract X consecutive video frames in the χ to form $F_n^{t_i}$:

$$F_n^{t_i} = (f_n^{t_i-X-1}, f_n^{t_i-X-2}, \dots, f_n^{t_i-1}, f_n^{t_i} | t_{min} + X \leq t_i \leq t_{max}). \quad (8)$$

$F_n^{t_i}$ is the conditional input, and $(F_n^{t_i}, h_n^{t_{con}})$ are combined to train our LDM. After the LDM has been trained, it can generate future potential haptic feedback $h_n^{t_{con}}$ from $F_n^{t_i}$.

IV. PROPOSED LDM-BASED HAPTIC TRANSMISSION ARCHITECTURE

In this section, we demonstrate the details of the proposed LDM-based haptic transmission architecture. It consists of three modules: 1). video frame extraction module; 2). haptic feedback generation module; and 3). haptic feedback generation acceleration module.

A. Video Frame Extraction Module

As shown in Fig. 1, the edge server receives video frames from the cloud server and uses the trained LDM to generate haptic feedback. According to Equation (7), it is crucial to select the appropriate frames based on a temporal dimension—frames that are neither too early, which can reduce the generation accuracy, nor too late, which may fail to meet the 1 ms latency requirement for haptic feedback.

Hence, the video frame extraction module selects the appropriate video frames by considering the accuracy of the generated haptic feedback and the network conditions. Specifically, when the network is under poor conditions, such as insufficient available bandwidth or high traffic load, it becomes necessary to increase the T_p at the expense of accuracy, ensuring that haptic feedback can still be transmitted within 1 ms. The impact of T_p in the accuracy of the generated haptic feedback is determined simulationally, and the simulation results are shown in Fig. 3.

B. Haptic Feedback Generation Module

This module is used to generate future potential haptic feedback that may occur using previous or current video frames as conditional information. Similar to the correspondence between words and pictures in a text-to-picture generation, the haptic feedback generation module corresponds video frames to future haptic feedback. Specifically, this module needs to correspond to the future haptic feedback $h_n^{t_{con}}$ to be generated with all the video frames in set χ .

We use contrast learning [17] to match video frames with their corresponding haptic feedback. The correct correspondence between a video frame and its haptic feedback is $(F_n^{t_i}, h_n^{t_{con}})$. Instead, the incorrect relationship is $(F_n^{t_i}, h_n^{t_{con'}})$. We train the video frame encoder $E_F(F_n^{t_i})$ with the haptic encoder $E_h(h_n^{t_{con}})$ by means of the following equation:

$$\mathcal{L}_{n,t_i}^{F,h} = -\log \frac{\exp(E_F(F_n^{t_i}) \cdot E_h(h_n^{t_{con}})/\tau)}{\sum_{j=1}^K \exp(E_F(F_n^{t_i}) \cdot E_h(h_n^{t_{con'}})/\tau)} \quad (9)$$

where τ is a small constant value and K is the total number of matching pairs $(F_n^{t_i}, h_n^{t_{con}})$. After training, matched pairs

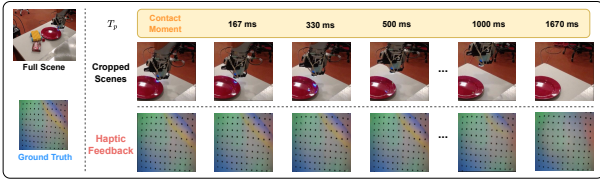


Fig. 2: An Example of generated haptic feedback.

(F_n^{ti}, h_n^{tcon}) have the biggest similarity after encoding, while mismatched pairs $(F_n^{ti}, h_{n'}^{tcon})$ have the least similarity after encoding. After obtaining the correspondence between video frames and haptic feedback, we can train our LDM using Equation (4) and generate future potential haptic feedback using Equation (5). As shown in Fig. 2, the trained LDM can generate the same haptic feedback from video frames at different timestamps that depict the same action. The cropped scene is intended to highlight the motion trend of the robotic arm, while the video frames used for generating haptic feedback remain in the full scene. However, when the T_p reached 1000 ms, the generated haptic feedback underwent significant changes, indicating that T_p has exceeded its maximum value $t_{con} - t_{min}$ at this point.

C. Haptic Feedback Generation Acceleration Module

According to Equation (6), a smaller haptic generation time l_g leads to a smaller T_p , enabling the use of video frames close to the moment of contact, which can improve the generation accuracy. Since l_g is around several hundred milliseconds level, which is much higher than other types of latency, it is crucial to minimize l_g . To minimize l_g , we employ an architecture, named DeepCache, proposed in [18]. The DeepCache can significantly reduce computational consumption during the sampling process to minimize l_g .

According to Equation (5), the feature z_n must pass through a denoising network ϵ_θ n times. At each step, ϵ_θ calculates the noise based on the input conditions $(z_n, E_C(\mathcal{P}), n)$. However, during successive steps, when the network processes $(z_n, E_C(\mathcal{P}), n)$ and $(z_{n-1}, E_C(\mathcal{P}), n-1)$ separately, it encounters highly similar features in certain layers of ϵ_θ . Caching these redundant features can prevent repeated computations, thereby significantly reducing the generation time l_g .

V. SIMULATION SETUP AND RESULTS

In this section, we first describe our simulation settings. Then we introduce the evaluation metrics, and we finally present and discuss the simulation results.

A. Simulation Settings

We use the *VisGel* dataset for our simulation [15]. It contains 12k touch actions, and each touch action consists of 300 pairs of temporally aligned video frames and haptic images as shown in Fig. 2. The video frames record a robotic arm containing haptic sensors moving towards and eventually touching the object being touched, the haptic is only generated after touching, and we use the pre-contact frames to predict the

haptic images. The haptic images record the deformation of the object touched by the haptic sensor, and the haptic images can be converted into contact area and normal force and thus into haptic feedback that can be perceived by humans [19]. We choose 80% of the dataset for training, while the remaining 20% for testing.

TABLE I: PARAMETER SETTINGS

Description	Value
Number of Frames for Condition Input	5
Learning Rate of Conditional Encoder	1×10^{-3}
Learning Rate of LDMs	2×10^{-6}
Sampling Steps	50
Bandwidth for 5G	5 Gbps
Bandwidth for Wi-Fi 6	1 Gbps
Traffic Load	0-100%
Packet Size of Haptic feedback	8 kbit
Processing Time of Haptic feedback	5 ms
Power of Local Rendering	13 W
Power of Generate Haptic	5 W
Energy Efficiency of Communications Equipment	0.1 W/Mbps
Power of Display Equipment	1 W
Data Rate (Local Rendering)	25 Mbps [2]
Data Rate (Cloud Rendering)	1 Mbps

We use the network structure shown in Fig. 1. For our simulations, users are connected to the access networks through 5G or Wi-Fi 6. The latency between servers and users fluctuates depending on the traffic load. We choose T_p ranging from 167 ms to 580 ms in the training process. In the testing phase, we use T_p ranges from 167 ms to 1670 ms. The ranges of T_p we used for testing are larger than for training. This is because, during training, too large the T_p will cause the robotic arm in the scene to be too far away from the object being touched. The video frame may contain insufficient information for generating the correct potential haptic feedback.

To evaluate the accuracy of our proposed haptic generation module, we choose three baseline approaches to generate potential haptic feedback. The specific methods are as follows:

- 1) PWA: Our proposed haptic generation module without acceleration module.
- 2) PGAN: We generate future potential haptic feedback using generative adversarial networks. We use the same video frames with haptic feedback pairs as in our proposed haptic generation module to train PGAN [15].
- 3) GLDM: We use LDM to generate current potential haptic feedback when the contact occurs, and the method generates haptic feedback only when the video of the moment of contact is received [16].

To evaluate the performance of our proposed architecture in haptic transmission, we choose five different scenarios, we abbreviate our proposed haptic feedback generation module as HFGM.

- 1) Cloud Server + HFGM + 5G: The generated future potential haptic feedback is transmitted from the cloud server, and the edge network adopts 5G.
- 2) Cloud Server + HFGM + Wi-Fi 6: The generated future potential haptic feedback is transmitted from the cloud server, and the edge network adopts Wi-Fi 6.

- 3) Edge Server + HFGM + 5G: The generated future potential haptic feedback is transmitted from the edge server, and the edge network adopts 5G.
- 4) Edge Server + HFGM + Wi-Fi 6: The generated future potential haptic feedback is transmitted from the edge server, and the edge network adopts Wi-Fi 6.
- 5) Edge Server + 5G: The haptic feedback is transmitted from the edge server, and the edge network adopts 5G.

We perform the simulation on a Linux server equipped with dual Intel Xeon Platinum 8352V CPU (36 cores and 72 threads, 2.1 GHz), 4 RTX 4090 (24 GB) GPU, 640 GB memory. Because LDM generates potential haptic feedback with some uncertainty, we perform 50 runs for each scene and calculate the average of the results to evaluate the haptic feedback generation accuracy. Details of the parameter settings are shown in Table I.

B. Evaluation Metrics

We use the quantitative metric Structural Similarity Index Measure (SSIM) [20] to evaluate the accuracy of generated haptic feedback. We use the transmission success rate p_{suc} to indicate the performance of our proposed architecture in different communication scenarios:

$$p_{suc} = n_{suc} / n_{sent} \quad (10)$$

where n_{sent} is the total number of haptic feedback sent, and n_{suc} is the number of haptic feedback that meets the latency requirement. Similarly, we calculate the average time t_{ave} for n_{sent} haptic feedback arrives at the users in advance as:

$$t_{ave} = (T_p - l_g - l_o - l_c) / n_{sent}. \quad (11)$$

We define the headset's energy consumption as follows:

$$E = e_{cmp} + e_{com}(d_l) + e_{dis} \quad (12)$$

where e_{cmp} is the computational energy consumption, including energy that is used for rendering videos and generating haptic feedback, and decoding energy. $e_{com}(d_l)$ is the energy used by the device for transmitting data, where d_l is the data load during the transmitting, and e_{dis} is the display energy consumption.

C. Simulation Results

We evaluate the generation accuracy of our haptic feedback generation module by measuring SSIM at different T_p that generates haptic feedback before the contact.

The quantitative results of generation accuracy on SSIM are shown in Fig. 3 (a). It shows that even though the action does not undergo a wide range of abrupt changes over 1000 ms, the generation still decreases with the increase of T_p , although the decrease remains within acceptable limits. However, when T_p beyond 1000 ms, the accuracy begins to decrease significantly as the T_p increases. Our proposed generation module has better generation accuracy compared to PWA and GLDM, and the acceleration module we proposed incurs almost no loss in generation accuracy compared to PWA. The generation time of haptic feedback is shown in Fig. 3 (b), where our proposed

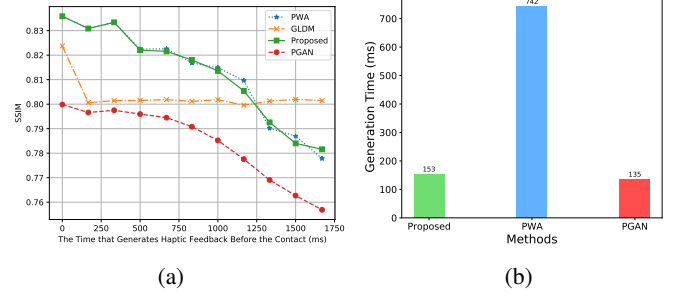


Fig. 3: Quantitative results for haptic feedback generation module and acceleration module: (a) Impacts of the time T_p that generates future potential haptic feedback before the contact on generated feedback accuracy. (b) The average time t_{ave} that haptic feedback arrives at the users in advance.

acceleration architecture reduces the generation time from 742 ms to 153 ms with almost no impact on the accuracy of the generation.

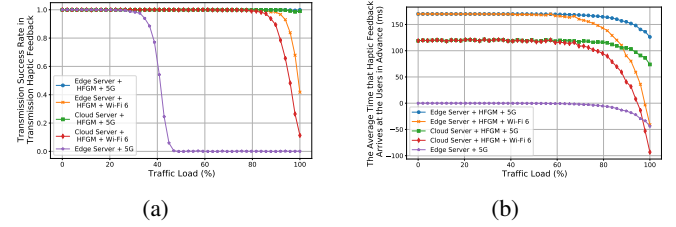


Fig. 4: Quantitative results for proposed transmission architecture: (a) Impacts of traffic load on the transmission success rate p_{suc} in transmission haptic feedback. (b) Impact of Traffic Load on Required Lead Time for Generated Haptic Feedback Delivery to Users.

Fig. 4 (a) illustrates the impact of traffic load on the transmission success rate p_{suc} of haptic feedback. Clearly, as the traffic load increases, each scenario experiences a decline in the p_{suc} . In the scenario Edge Server + 5G, there is a significant drop in the p_{suc} when the traffic load reaches just 35%. However, when our haptic feedback generation module is employed to generate future potential haptic feedback, the p_{suc} remains stable until the traffic load exceeds 80%, even in the Cloud Server + HFGM + Wi-Fi 6. In contrast, if 5G is used as the access network, in the Cloud Server + HFGM + 5G and Edge Server + HFGM + 5G, the p_{suc} stays around 100% for transmissions from either cloud or edge server, with only a slight decrease of no more than 2% when the traffic load approaches 100%. Fig. 4 (b) shows the impact of traffic load on the time t_{ave} which haptic feedback arrives in advance to the users. When the HFGM is used to generate future potential haptic feedback, the feedback reaches the users before contact occurs, regardless of the scenario. Furthermore, as the traffic load increases, transmission latency also increases, resulting in a reduction in the t_{ave} of haptic feedback arrival across all scenarios. Specifically, in the Edge Server + HFGM + 5G,

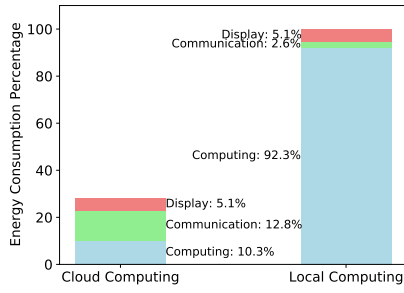


Fig. 5: Estimation of user-device energy consumption for different computing methods.

t_{ave} can reach 170 ms. Even at the highest network load, the potential haptic feedback can still be transmitted nearly 100 ms in advance to the users.

The power consumption of the Proposed architecture for cloud computing and local computing is shown in Fig. 5. For local computing, approximately 94% of the total energy is consumed by computing tasks, whereas in cloud computing, computing accounts for only 10.3% of total energy consumption since the headset does not perform local computations. Overall, the headset can obtain a 71% reduction in total energy consumption when using cloud computing compared to local computing. The lifespan T_l can be expressed by the formula:

$$T_l = C/E \quad (13)$$

where C represents the battery capacity. When the total energy consumption is reduced by 71% the headset can achieve the same lifespan with only one-third of the battery capacity.

VI. CONCLUSION

To address the 1 ms latency challenge that is caused by moving computing tasks to the cloud for rendering, we propose a generative AI-based transmission architecture. The architecture uses LDM to generate future potential haptic feedback based on the current or previous video frames and transmits the generated haptic feedback to the users in advance to meet the haptic latency requirement.

For future work, one of the future directions is to utilize video frames that are closer to the moment of contact to address scenarios involving fast-moving objects in the scene.

VII. ACKNOWLEDGMENT

This work is partially funded by the National Natural Science Foundation of China under Grant 62201277 and 62071255, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20220390, in part by Key R and D Program of Jiangsu Province Key project and topics under Grant (BE2023035), in part by the Brazilian National Council for Scientific and Technological Development - CNPq, via Grant No.306607/2023-9.

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